

Intro

$x_{k+1} = f_k(x_k, u_k, w_k) \quad k = 0, 1, \dots, N-1$
 w_k r.v. cond indep with $x_l, u_l, w_l, \quad l < k$ given x_k, u_k
 and $p(w_k|x_k, u_k)$

In general cost function is $\mathbb{E}_{(X_1, U_0, W_0|x_0)}[g_N(x_N) + \sum_{k=0}^{N-1} g_k(x_k, u_k, w_k)]$

OPEN loop fixed $\bar{u}_0, \dots, \bar{u}_{N-1}$ has $\prod_{k=0}^{N-1} N_u^k$ strats.

CLOSED loop $u_k = \mu_k(x_k)$, admiss policy is $\pi = (\mu_0(\cdot), \dots, \mu_{N-1}(\cdot))$. Expected closed loop cost ass with adm pol $\forall x \in S_0$ π is $J_\pi(x) = \mathbb{E}_{(X_1, W_0|x_0=x)}[g_N(x_N) + \sum_{k=0}^{N-1} g_k(x_k, \mu_k(x_k), w_k)]$. Optimal cost is $J^*(x) = J_{\pi^*}(x)$. Total strategies $\prod_{k=0}^{N-1} (N_u^k)^{N_x^k}$

Discr, Finite state conveniet express dynamics as $P_{ij}(u, k)$ and say $x_{k+1} = w_k$ where $p_{w_k|x_k, u_k}(j|i, u) = P_{ij}(u, k)$. Conversely given $x_{k+1} = f(\cdot)$ and $p_{w_k|x_k, u_k}$ u get $P_{ij}(u, k) = \sum_{\bar{w}_k s.t. f_k(i, u, \bar{w}_k)=j} p_{w_k|x_k, u_k}(\bar{w}_k|i, u)$

DPA

Goal $\pi^* = \argmin_{\pi} J_\pi(x) \quad \forall x \in S_0$

Prc of Opt let π^* , cons problem at $x \in S_i$, then $\pi_i^* = (\mu_i^*(\cdot), \dots, \mu_{N-1}^*(\cdot))$ is optimal.

DPA $\forall x \in S_0, \quad J^*(x) = J_0(x)$ given by: 1- $J_N = g_N(x) \forall x \in S_N$ 2- $J_k(x) = \min_{u \in U_k(x)} \mathbb{E}_{(w_k|x_k=x, u_k=u)} [g_k(x_k, u_j, w_k) + J_{k+1}(f_k(\dots))] \forall x \in S_k, k = N-1, \dots, 0$ and if u take u_k that minim J_k u get an optim policy. Total operts is $\sum_{k=0}^{N-1} N_x^k N_u^k + N_u^N$

Time Lags if $x_{k+1} = f_k(x_k, x_{k-1}, u_k, u_{k-1}, w_k)$ u augment the state $\tilde{x}_k = (x_k, y_k, s_k)$ where $y_k = x_{k-1} \quad s_k = u_{k-1}$ so augment $\tilde{f}_k = (f_k(x_k, y_k, u_k, s_k, w_k), x_k, u_k)$

Correlated dist $w_k = C_k y_{k+1} \quad y_{k+1} = A_k y_k + v_k \quad x_{k+1} = f_k(x_k, u_k, w_k)$. Augument $\tilde{x}_{k+1} = [x_{k+1}, y_{k+1}]$

Forecasts distr of w_k known when given $y_{k+1} = v_k$. Augment $\tilde{x}_k = (x_k, y_k)$ and $\tilde{w}_k = (w_k, v_k)$ so $p(\tilde{w}_k|\tilde{x}_k, u_k) = p(w_k|y_k)p(v_k)$

Infinite Horizon

Consider **time invariant** setup (exopt for μ , show later). Write DPA with $V_l(\cdot) = J_{N-l}(\cdot)$ assuming V_l conv u get **Bellman Equation** $J(x) = \min_{u \in U(x)} \mathbb{E}_{(w|x=x, u=u)} [g(x, u, w) + J(f(x, u, w))]$ J is optimal. Diff solve $x \in S$ simult.

Infinite Horizon

SSP Finite state, time inv, trans prob. time inv (general setup are not time inv). **Assume** $\exists 0 | P_{00}(u) = 1, g(0, u, 0) = 0$ and $\exists$ at least 1 prop policy (proper = P end in $0 > 0$) and all others have i s.t. $J(i) = \infty$. **Theorem** $\forall V_0(1), \dots, V_0(n)$, generate $V_{l+1}(i) = \min_{u \in U(i)} (q(i, u) + \sum_{j=1}^n P_{ij}(u) V_l(j)) \forall i \in S^+$ where $q(i, u) = \mathbb{E}[g(x, u, w)]$ convergs to $J^*(i) \forall i \in S^+$. Sats BE, sol. is unique and gives opt policy.

LQR

Quadr stage cost $g(x, u) = qx^2 + ru^2$, lin dyn $f(x, u) = ax + bu$. Educ guess $J(x) = px^2$. If $b = 0 \quad p = \frac{q}{1-a^2}$ req $|a| < 1$. If $b \neq 0$ let $\gamma = r(1-a^2) - b^2q$ then $p = \frac{-\gamma + \sqrt{\gamma^2 + 4b^2qr}}{2b^2}$

Solving BE

VI apply DP recursion above. Requires ∞ iterations.

PI Under Ass1, $\forall$ prop μ the assoc $J_\mu = (J_\mu(1), \dots)$ is uniq sol of $J_\mu(i) = q(i, \mu(i)) + \sum_j P_{ij}(\mu(i)) J_\mu(j)$ (set of lin eqts). Sequence $V_{l+1} = (J_\mu(1), \dots)$ is uniq sol of $J_\mu(i) = q(i, \mu(i)) + \sum_j P_{ij}(\mu(i)) J_\mu(j)$ or you can do iterations: $V_{l+1}(i) = q(i, \mu(i)) + \sum_j P_{ij}(\mu(i)) V_l(j)$

PI Algo 1: P Eval $J_{\mu^h}(i) = q(i, \mu^h(i)) + \sum_j P_{ij}(\mu^h(i)) J_{\mu^h}(j)$ 2: Impr $\mu^{h+1}(i) = \argmin_{u \in U(i)} (q(i, u) + \sum_j P_{ij}(u) J_{\mu^h}(j))$

Reprehase VI Stage 1: $V_{l+1}(i) = q(i, \mu^l(i)) + \sum_j P_{ij}(\mu^l(i)) V_l(j)$ stage 2 $\mu^{l+1}(i) = \argmin_u (q(i, u) + \sum_j P_{ij}(u) V_{l+1}(j))$ **Complex PI** stage 1 $O(n^3)$, stage 2 $O(n^2p)$. Total $O(n^2(n+p))$ at each it.

Compl VI $O(n^2p)$ at each it.

Gass Seidal Upd for VI: use most recent $V(j)$ every time u can. Faster conv, less memory.

Linear Algebra Stage 1 PI is $J = G + PJ$ has uniq sol iff P not have eignvl = 1 iff policy is proper

Linear Programming

Solution of maximize $V \sum_{i \in S^+} V(i)$ subj to $V(i) \leq (q(i, u) + \sum_{j=1}^n P_{ij}(u) V(j)) \quad \forall u \in U(i) \quad \forall i \in S^+$ also solves BE of the SSP.

Rewrite in can form minim in $x \quad c^T x$ subj to $Ax \leq b$. Do $x = [V(1), \dots, V(n)]^T \quad c = [-1, \dots, -1]^T$ and $A = [I - \alpha P(u_1), \dots, I - \alpha P(u_m)]^T$ and $b = [q(1, u_1), \dots, q(n, u_1), \dots]^T$

Discounted Problems

No termin state, disc factor, no need for assmpt 1 and 2. Now $J_\pi(i) = \mathbb{E}_{(X_1, W_0|x_0=i)} [\sum_{k=0}^{N-1} \alpha^k g(x_k, \mu_k(x_k), w_k)]$. **Th** Opt cost to go sat BE with disc factor, unique, gives opt pol.

Aux SSP equiv to disc: introduce 0 and $\tilde{U}(0) = \text{stay}$. Dinmc $\tilde{x}_{k+1} = \tilde{w}_k$ with $\tilde{P}_{ij}(u) = \alpha P_{ij}(u) \forall i, j \in S^+$, $\tilde{P}_{i0}(u) = 1 - \alpha \forall i \in S^+$ and $\tilde{P}_{0j}(u) = 0 \forall j \in S^+$ and $\tilde{P}_{00}(u) = 1$ and costs $\tilde{g}(\tilde{x}_k, \tilde{u}_k, \tilde{w}_k) = \alpha^{-1} g(\tilde{x}_k, \tilde{u}_k, \tilde{w}_k)$ and $\tilde{g}(\tilde{x}_k, \tilde{u}_k, 0) = 0$.

Disc problms conv to opt cost to go geometricly with exp factor α , smaller $\alpha \rightarrow$ faster conv of VI

SSP and Det finite state systms

Ass $\forall i \in \mathcal{V} \quad \forall Q \in \mathbb{Q}_{i,i} \quad J(Q) \geq 0$ no neg cycls.

DFSS assume no dist and finite S_k .

DFS to SP $\mathcal{V} = U_k \{(k, x_k) | x_k \in S_k\} \cup \{T\}$, define $c(i, j) = g_N(x_N)$ if $i = (N, x_N), j = T$ and $c(i, j) = \min_{u \dots} g_k(x_k, u)$ if $i = (N, x_k), j = (k+1, x_{k+1})$ and $c(i, j) = +\infty$ otherwise

SP to DFS Introduces $c_{i,i} = 0$ so optim path has $|\mathcal{V}|$. Let $N = |\mathcal{V}| - 1, S_k = \mathcal{V} \setminus \{T\} \quad k = 1, \dots, N-1$ and $S_N = \{T\}$ and $S_0 = \{s\}$. Then $U_k = \mathcal{V} \setminus \{T\} \quad k = 0, \dots, N-2$ and $U_{N-1} = \{T\}$. Let $x_{k+1} = u_k, g(x_k, u_k) = c_{x_k, u_k} \quad k = 0, \dots, N-1$ and $g_N(T) = 0$. Solve DFS via DPA.

Hidden MM Let $M_{ij}(z) = p_{z|x,w}(z|i, j) \quad z \in \mathcal{Z}$ meas. space. Given measurmts $Z_1 = (z_1, \dots, z_N)$ find $\hat{X}_0 = \argmax_{X_0} p(X_0|Z_1) = \argmax_{X_0} p(X_0, Z_1)$. Find $p(X_0, Z_1) = p(x_0) \prod_{k=1}^N P_{x_{k-1}x_k} M_{x_{k-1}x_k}(z_k)$ Maximize equiv to minimize $X_0 (c_{S,(0,x_0)} + \sum_{k=1}^N c_{(k-1,x_{k-1}),(k,x_k)})$. Where $c_{S,(0,x_0)} = -\ln(p(x_0))$ and $c_{(k-1,x_{k-1}),(k,x_k)} = -\ln(P_{x_{k-1}x_k} M_{x_{k-1}x_k}(z_k))$ if prod > 0

SSP Algos

LCA 0- s in OPEN, $d_s = 0, d_j) \infty$ 1 rmv i from OPEN, exct 2 for all childrens j of i 2 if $d_i + c_{i,j} < d_j$ and $d_i + c_{i,j} < d_T$ set $d_j = d_i + c_{i,j}$. If $j \neq T$ put j in OPEN 3 if OPEN empty end, else go to 1.

Th if path $\exists$ then LCA termtes $d_T = J_{Q^*}$, othwise $d_T = \infty$.

Depth first LIFO. Breadth FIFO. Best First remove i^* s.t. $d_{i^*} = \min_{i \in \text{OPEN}} d_i$. **A*** to enter OPEN need $d_i + c_{i,j} + h_j < d_T$

Deterministic Cont Time control . HJB Eq

$$\frac{d(F \circ G)(t)}{dt} = \frac{dF(G(t))}{dt} = \frac{dF(x)}{dt} \Big|_{x=G(t)} \frac{dG(t)}{dt}.$$

$$F(x, y) \rightarrow \Delta F \approx \frac{\partial F(x, y)}{\partial x} \Delta x + \frac{\partial F(x, y)}{\partial y} \Delta y$$

$$F(x(t), y(t)) \rightarrow \Delta F \approx \frac{\partial F(x, y)}{\partial x} \frac{dx(t)}{dt} \Delta t + \frac{\partial F(x, y)}{\partial y} \frac{dy(t)}{dt} \Delta t$$

$$\text{If vector field } \Delta F = [\Delta F_1 \dots \Delta F_p]^T \approx J_F[\Delta x_1 \dots \Delta x_n]$$

where in $J \in \mathbb{R}^{p \times n}$

$$\frac{\partial F(G(x, u), y)}{\partial x} = \frac{\partial F(z, y)}{\partial z} \Big|_{z=G(x, u)} \frac{\partial G(x, u)}{\partial x}$$

$$\frac{\partial F(t, x, \mu(x, t))}{\partial t} = \frac{\partial F(t, x, u)}{\partial t} \Big|_{u=\mu(x, t)} +$$

$$\frac{\partial F(t, x, u)}{\partial u} \Big|_{u=\mu(t, x)} \frac{\partial \mu(t, x)}{\partial t}$$

$$\frac{dV(t, x(t))}{dt} = \frac{\partial V(t, x)}{\partial t} \Big|_{x=x(t)} \frac{dt}{dt} + \frac{\partial V(t, x)}{\partial x} \Big|_{x=x(t)} \frac{dx(t)}{dt}$$

$$\Delta V = V(t + \Delta t, x(t + \Delta t)) - V(t, x(t)) = \frac{dV(t, x(t))}{dt} \Delta t + o(\Delta t)$$

$$\text{Dynamics } \dot{x}(t) = f(x(t), u(t))$$

$$\text{Cost } h(x(T)) + \int_0^T g(x(\tau), u(\tau)) d\tau, \text{ cost of feedback}$$

$$\text{law is } J_\mu(t, x) = h(x(T)) + \int_t^T g(x(\tau), u(\tau)) d\tau$$

Ass $\forall$ admis , init t and in cond $\mu \exists ! x(\tau)$ that sat dynamics

$$\text{HJB } 0 = \min_u [g(x, u) + \frac{\partial J^*(t, x)}{\partial t} +$$

$$\frac{\partial J^*(t, x)}{\partial x} f(x, u)] \quad \forall t \in [0, T] \quad \forall x \in S \text{ subj to } J^*(T, x) = h(x) \text{ suff condit for opt}$$

Pontryagin's Min Princ

If $F(t, x, u)$ cont diff, assume $\exists \mu^*(t, x) \in \arg\min_u F(t, x, u)$ then $\forall t, x \quad \frac{\partial (\min_u F(t, x, u))}{\partial t} =$

$$\frac{\partial F(t, x, u)}{\partial t} \Big|_{u=\mu^*} \text{ and } \frac{\partial (\min_u F(t, x, u))}{\partial x} = \frac{\partial F(t, x, u)}{\partial x} \Big|_{u=\mu^*}$$

Min Princ Given $x(0) = x \in S$, let $u(t)$ opt contr traj assoc with $x(t)$. Then $\exists p(t)$ s.t.

$$\dot{p}(t) = - \frac{\partial H(x, u, p)}{\partial x} \Big|_{x(t), u(t), p(t)}^T \text{ and } p(T) = \frac{\partial h(x)}{\partial x} \Big|_{x(T)}^T \text{ and}$$

$$u(t) \in \arg\min_u H(x(t), u, p(t)) \text{ and } H(x(t), u(t), p(t)) = \text{const } \forall t \in [0, T] \text{ where } H(x, u, p) = g(x, u) + p^T f(x, u).$$

If $u(t)$ sat those condts can be optimal.

MP is necess cond for optim, while HKB is suff (if sat HJB $\rightarrow$ optim) , (if cand sol is diff, it must sat HJB)

Fixed Terminal State Supp $x(0) = x_0$ and $x(T) = x_T$, then $h(x(T))$ not useful, thus $J(T, x)$

not diff and **not** hold $p(T) = \frac{\partial h(x)}{\partial x} \Big|_{x(T)}^T$. But we have $2n$ ODEs and $2n$ BCs. If $x_i(T) = x_{T,i}$ only for some i 's.

Then for the other coord u put $p_j(T) = \frac{\partial h(x)}{\partial x_j} \Big|_{x(T)}$

Free Initial State

$x(0)$ free but extra $l(x(0))$ so tot cost is $l(x(0)) + J(0, x(0))$. Nec cond is $(\frac{\partial l(x)}{\partial x} + \frac{\partial J(0, x)}{\partial x}) \Big|_{x_0} = 0$ so we

can get the adjoint BC $p(0) = - \frac{\partial l(x)}{\partial x} \Big|_{x(0)}^T$

Free Terminal State Compute opt traj for various T and look for best T . So optimal T must sat

$$\frac{\partial J(t, x)}{\partial t} \Big|_{t=0, x(0)} = 0, \text{ we get that } H(x(t), u(t), p(t)) = 0 \forall t \in [0, T]$$

Time Varying System and Cost

System and stage cost vary with time

1st order ODEs

Separable $y' = a(x) b(y)$

$$\frac{dy}{b(y)} = a(x) dx \Rightarrow \int \frac{1}{b(y)} dy = \int a(x) dx + C$$

Autonoma $y' = f(y)$

$$\int \frac{1}{f(y)} dy = x + C$$

Lineare $y' + p(x)y = q(x)$

Fattore integrante: $\mu(x) = e^{\int p(x) dx}$

$$(\mu y)' = \mu q \Rightarrow y(x) = \mu(x)^{-1} \left(\int \mu(x) q(x) dx + C \right)$$

Omogenea (in y/x) $y' = F\left(\frac{y}{x}\right)$

Sost: $y = vx \Rightarrow y' = v + xv'$

$$v + xv' = F(v) \Rightarrow \frac{dv}{F(v) - v} = \frac{dx}{x}$$

Bernoulli $y' + p(x)y = q(x) y^n, \quad n \neq 0, 1$

Sost: $z = y^{1-n} \Rightarrow z' + (1-n)p(x)z = (1-n)q(x)$

(poi risolvi come lineare in z)

Esatta $M(x, y) dx + N(x, y) dy = 0$

Cond: $\partial_y M = \partial_x N \Rightarrow \exists \Phi : d\Phi = M dx + N dy$

Sol: $\Phi(x, y) = C$

Integrating factor (se non esatta):

$$\text{se } \frac{\partial_y M - \partial_x N}{N} = f(x) \text{ allora } \mu(x) = e^{\int f(x) dx}$$

$$\text{se } \frac{\partial_x N - \partial_y M}{M} = g(y) \text{ allora } \mu(y) = e^{\int g(y) dy}$$

Riccati (cenno) $y' = a(x)y^2 + b(x)y + c(x)$

Se conosci una sol. particolare y_p : sost $y = y_p + \frac{1}{v}$

$\Rightarrow$ EDO lineare in v .